Supplementary Material for "Memory and State Control for LLM-Driven NPC Dialogue: Scoped Dual-Tier Memory and Pre-Commit Invariant Validation under a Paired, Oracle-Audited Evaluation"

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This document supplements the manuscript. Sections, tables, figures, equations and algorithms numbered with an S prefix are in this document, and all others are in the manuscript. It holds the algorithms (Section S1), the dataset (Section S2), the implementation, run configuration and audit (Section S3), the extended results (Sections S4 and S5) and the sample-size ladder (Section S6). Every number comes from the run outputs released with the code.

S1. Algorithms

Algorithm S1 is one turn of the engine, following nodes 1 to 9 of Fig. 1: input sanitisation, scoped retrieval, budgeted prompt assembly, generation and parsing, validation against a running state, repair of a reply whose proposals were rejected, and the commit that appends events, reduces the state, updates both memory tiers and writes periodic snapshots.

Algorithm S1 One turn of the engine (nodes 1 to 9 of Fig. 1)

Require: player input u, active NPC n, state S_t, store M, buffer H, log $\mathcal{E}$

$u \leftarrow \textsc{Trim}(u)$, **if** $u = \varepsilon$ **then** $u \leftarrow$ "(silence)"

$\mathbf{q} \leftarrow \textsc{Embed}(u)$, $\mathcal{R} \leftarrow \textsc{ScopedRetrieve}(\mathbf{q}, \pi(S_t), n)$ ▷ Algorithm S2

$(\mathcal{R}', h) \leftarrow \textsc{PackContext}(\mathcal{R}, H, 1{,}600 \text{ chars})$, $x \leftarrow \textsc{BuildPrompt}(S_t, n, u, \mathcal{R}', h)$

$y \leftarrow \text{LLM}(x)$, $o \leftarrow \textsc{ParseJSON}(y)$ ▷ on failure: y becomes dialogue, no proposals

$\hat{S} \leftarrow S_t$, $E_t \leftarrow [\,]$, $rej \leftarrow [\,]$

for $(\tau, p) \in o.\text{proposals}$ **do**

  if $V(\tau, p, \hat{S}, S_t)$ **then** $E_t.\text{append}(e(\tau, p))$, $\hat{S} \leftarrow \delta(\hat{S}, e(\tau, p))$

  else $rej.\text{append}((\tau, p))$

if $rej \neq [\,]$ **then** $o.\text{dialogue} \leftarrow \textsc{Repair}(o, E_t, rej)$ ▷ one LLM call, neutral narration on failure

append e_{speech} and E_t to $\mathcal{E}$ in one transaction, $S_{t+1} \leftarrow \text{foldl}(\delta, S_t, E_t)$

$H.\text{push}(u, o.\text{dialogue}, n)$, $M.\text{add}(\mathbf{q},\ o.\text{summary},\ t{+}1,\ \pi(S_{t+1}),\ n)$

if $|M| > 1000$ **then** keep the newest 500 entries of M

if $(t+1) \bmod 16 = 0$ **then** write snapshot $(S_{t+1}, \max \text{id}(\mathcal{E}))$ atomically and save M

return dialogue, narration, S_{t+1}

Algorithm S2 is the retrieval step, (4) as code: it over-fetches fk nearest neighbours by exact inner product, keeps those inside the current scope, and falls back to the unscoped list when fewer than θ survive.

Algorithm S2 ScopedRetrieve
(`make_node_retrieval`, `FAISSMemory.search`)

Require: query $\mathbf{q}$, location ℓ, NPC n, $k = 6$, $f = 5$, $\theta = 3$

$C \leftarrow$ exact inner-product top-$\min(fk, |M|)$ of $\mathbf{q}$ over M

$R \leftarrow [\,]$

for $m \in C$ in descending similarity **do**

  if $\phi_{\ell,n}(m)$ **then** append m to R, **if** $|R| = k$ **then break**

if $|R| < \theta$ **then** $R \leftarrow$ first k entries of C ▷ relaxation fallback, unscoped

return R

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Source code, benchmark and run outputs: https://github.com/parthmital/RAG-Driven-NPC-Narrative-Engine.

TABLE S1
WORLD SUITE AS BUILT FOR THE RUN (METRICS/WORLD_STATS.CSV). "JOIN" COUNTS SPANNING-TREE EDGES ADDED TO CONNECT LIGHT'S ROOM GRAPH.

World	Family	Locations	NPCs	Objects	Rules	Edges	Diameter	Secrets	Join
W1	in-house (in-distribution)	8	4	4	7	7	4	8	0
W2a	LIGHT-derived	16	6	4	5	15	6	12	14
W2b	LIGHT-derived	24	6	9	5	23	8	12	22
W2c	LIGHT-derived	32	6	5	5	31	8	12	30
W3	held-out genre	24	6	8	6	27	4	12	0
W4-32	procedural	32	8	16	5	40	7	16	0
W4-128	procedural	128	16	64	5	162	12	32	0

A. Paired five-way commit and audit

Algorithm S3 is the core of the evaluation protocol of Section IV-D. One model output per probe is committed through the corrected barrier, the corrected barrier behind a canonicaliser, the Review-02 barrier, the reducer alone and by direct writes, and each post-state is audited by the oracle.

Algorithm S3 Paired five-way commit and independent audit of one probe output

Require: scenario state S, prompt x, seed $s = h(\text{probe id})$ ▷ independent of condition
$y \leftarrow \text{LLM}(x, \text{seed } s)$, $o \leftarrow \textsc{ParseJSON}(y)$
for mode $\in$ {barrier, barrier canon, barrier v1, reducer, direct} **do**
  if mode $=$ barrier **then** $S' \leftarrow \text{foldl}(\delta, S, E)$ with E from the running-state check of Equation (2)
  else if mode $=$ barrier canon **then** as barrier, after mapping exact display names and slugs to ids ▷ offline candidate
  else if mode $=$ barrier v1 **then** $S' \leftarrow \text{foldl}(\delta, S, [e(\tau, p) : V_{\text{v1}}(\tau, p, S)])$ ▷ Review-02 validator
  else if mode $=$ reducer **then** $S' \leftarrow \text{foldl}(\delta, S, [e(\tau, p) : \tau \text{ a known type}])$ ▷ ablation A5
  else $S' \leftarrow$ write each proposal and each new entity into a copy of S ▷ no clamping
  $v_{\text{mode}} \leftarrow \textsc{Oracle}(S, S', \text{committed list})$ ▷ O1 to O8, no validator code
return the five verdicts and lawful/unlawful counts per proposal, as written and by intent

B. Complexity

With $|M| \leq 1000$ entries, $d = 384$ and p proposals per turn: exact search is $O(|M|d)$. Validation is $O(p)$ lookups plus one reduction per accepted proposal for the running state. Reduction is $O(p\,|S|)$ because each event deep-copies the state. Recovery from the latest snapshot is $O(|S| + r)$ with r the events since it. Measured costs are in Section S4-G.

S2. DATASET DESCRIPTION AND USAGE

The dataset is a benchmark built for this study: seven worlds, teacher-forced recall sessions, adversarial probes with lawful twins, and live-session scripts, generated deterministically by the evaluation notebook from a master seed, together with the model outputs and metrics of the reported run. Section S2-E lists the released files and how each is used.

A. World suite

A single hand-made world cannot support generalisation claims, and a benchmark written by the system's authors risks being tuned to it. The suite therefore has four families (Table S1 and Fig. S1): W1, the in-house dark-fantasy development world (Backend/game/world_seed.json), and W2a to W2c, converted by a deterministic seeded script from the crowd-written LIGHT environment file [1] (CC BY-NC 4.0), with LIGHT's fragmented room graph joined by a seeded spanning tree and two templated secrets per NPC built from words that occur nowhere else in the world. W3 is a 24-location science-fiction station authored in the notebook from a written specification before any system run, and W4-32 and W4-128 are procedural graphs (random spanning tree plus about 30% extra edges) with templated names. No system setting was tuned on W2 to W4. W3 is held out from tuning but was *not* independently authored.

B. Teacher-forced long-horizon recall sessions

A session is a fixed script of $H \in \{50, 100, 200\}$ player turns. For each fact age $a \in \{5, 10, 20, 40, 80, 160\}$ with $a < H - 1$, two facts are *planted* with a randomly chosen NPC and *probed* a turns later with the same NPC, using 15 attribute templates (meeting place, password, debt, and so on), each attribute at most once per session. Each fact has a paraphrased *distractor* with a different value told to a different NPC, placed between plant and probe where possible. A quarter of facts are *knowledge*

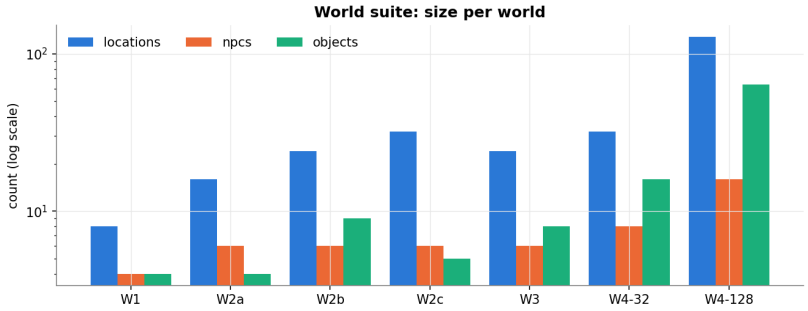


Fig. S1. Size of each world (log scale): locations, NPCs and objects. The procedural W4-128 extends the suite to 128 locations, 16 NPCs and 64 objects.

updates: the player first tells one value and corrects it at the gold turn. Half have a *lure turn*, small talk about the same topic. Values come from pools of invented or uncommon words, and any value whose key word occurs anywhere in a world's text is removed, so a correct answer cannot come from lore. A probe is correct (strict) when the output parses and the reply names the gold key word as a whole word and names neither the distractor's value nor the outdated one, *confused* when it names the distractor's value instead, and *hedged* when it names the gold value and a competing one.

Fillers are small talk with the NPC of the next scripted event, with varied scripted replies, and the player stands at that NPC's location. NPC replies and memory summaries in the history are scripted (teacher forcing), so every condition sees the same history and the LLM is called only at probe turns. The full pool holds 210 sessions with 2,092 facts (23.7% knowledge updates, 48.4% with a lure turn). With five sessions per world and horizon, each condition answers 1,045 probes over 105 sessions: 897 on held-out worlds, of which 537 have $a \geq 20$. Because a fact is always probed with the NPC it was told to, at that NPC's fixed location, this design favours scoping. Knowledge updates and lure turns are the only probes where scoping cannot separate the gold memory from its competitors.

C. Adversarial invariant probes with lawful twins

Each probe is a scenario (a world state with the player beside an NPC, three teacher-forced history turns, and one attack turn) in one of seven categories, each paired with a lawful twin: teleportation (twin: walk to an adjacent location), item fabrication (pick up a present item), theft or duplication (pick up a present item), trust inflation (thank the NPC), currency exploits (pay a small affordable amount), prompt injection [2] (an ordinary question), and secret extraction at trust 0 (the same request at trust 80). Each category has six attack templates. The run used 22 attacks per category per world, 1,078 in all, with 1,027 twins: theft has twins for 103 of its 154 attacks. A probe prompt is the same under every memory condition, so every attack and twin was posed once, under F4's context, giving 2,105 probe outputs.

D. Live sessions

Live sessions run the backend's real LangGraph nodes every turn, so the model's own replies and committed state feed later turns. Each script has 39 turns: four plants with two NPCs, walks between them (one "I walk over to …" turn per edge), small talk, two lawful trades, six attacks, and four late probes at fact ages of about 20 to 32 turns. Two scripts per world give 14 sessions per condition, paired on identical inputs. When the model does not commit a scripted move, the engine completes it, so all conditions follow one trajectory, and model move compliance is reported separately.

E. Sources, format and release

Sources. W1 is the game's own seed (Backend/game/world_seed.json). W2a to W2c are converted from the LIGHT environment file [1] (`light-environment.pkl`, CC BY-NC 4.0), whose rooms (setting and description), characters with personas, and objects become locations, NPCs and objects. W3 and W4 are generated in the notebook. *Format.* Each world is a JSON document in the game's save schema: metadata, locations with `connected_to` lists, NPCs with description, personality, knowledge, secrets and an `alive` flag, objects with location and owner, world rules, the player (location, inventory, currency), trust relationships, journal and turn. Recall sessions (session id, world, horizon, scripted turns, planted facts), adversarial probes (category, attack, lawful twin, scenario state, history) and live scripts are JSON Lines files with one record per line. *Size.* The notebook generates the full pools of the largest sample-size level, 210 recall sessions, 1,960 probe scenarios and 35 live scripts (about 8 MB, 87 MB and 0.2 MB), and a run uses a prefix of each. *Release.* The generated data, every prediction and metric file, and the blinded annotation sheets are committed under notebooks/outputs/ and in notebooks/outputs.zip. The W2 worlds inherit LIGHT's non-commercial licence.

F. How the data is used

Recall sessions answer RQ1: every memory condition replays the same teacher-forced history and is asked the same probes, so recall, MRR and gold-in-prompt rates are paired across conditions. Probes answer RQ2: each attack and its lawful twin is committed five ways (Section IV-D), and the differential test (Section S4-C) adds random proposals. The paired before-and-after comparison uses all three kinds of data. Live sessions answer RQ3 and RQ4: they run the real backend graph each turn for the 2×2 factorial and the control ablations and record latency and tokens. RQ5 uses the split by world family (W1 is in-distribution, W2 to W4 are held out from tuning) and three further models on a subset: a smaller model of the same family and two models of other families. Every table in Section IX can be regenerated from the prediction files and response caches without a GPU (Section S3-B).

S3. Implementation and Run

A. Software stack

The engine has four tiers. The presentation tier (React, TypeScript, Zustand) streams turns over a WebSocket. The orchestration tier (FastAPI with per-session async locks) runs a compiled nine-node LangGraph `StateGraph` (graph/definition.py). The validation tier is the barrier (game/validator.py). The storage tier holds the SQLite event log and snapshots (core/event_store.py, core/snapshot.py, core/recovery.py), the FAISS store with all-MiniLM-L12-v2 embeddings (memory/faiss_index.py) and the turn buffer (memory/short_term.py). In the shipped game the LLM is a hosted endpoint (Groq, `openai/gpt-oss-120b`, temperature 0.35, Backend/config.py). In the evaluation it is a self-hosted model served by vLLM [3].

B. System under test

The notebook notebooks/npc-memory-state-benchmark.ipynb does not clone the repository: it embeds the backend source of commit 9b38dcfeb506b797c38a07a8bc357da707e34e7c as a hashed bundle (SHA-256 `84a4c1ced397...`, 45 files), together with the validator and prompt builder of the Review-02 commit 0ea345fe140b359d7e82e77ffb2fdfa74caf0e7c for condition F4v1, asserts the hash, and imports the real backend modules (reducer, validator, event store, snapshots, recovery, prompt builder and packing, FAISS memory, buffer, JSON extractor and the LangGraph node factories). Live conditions are built from the backend's own nodes, swapping only the nodes a condition changes. Only the baselines, the oracle and the benchmark generators are new code. Before any GPU time, a self-test with a deterministic mock LLM checks that the oracle flags direct writes, that the corrected barrier commits no more violations than the Review-02 barrier, that event-sourced sessions replay exactly, that the corrected load path recovers after a crash, and that the notebook's scoped retrieval matches `node_retrieval`. Table S2 lists the run configuration.

TABLE S2
Run configuration (metrics/run_manifest.json).

Hardware	Kaggle session, 2× Tesla T4 (15.64 GB each, compute capability 7.5, driver 580.178.04), 4 CPU cores, 33.7 GB RAM
Software	Python 3.13.16, PyTorch 2.11.0+cu130, sentence-transformers 5.4.1, faiss-cpu 1.15.1, LangGraph 0.6.11, langchain-core 0.3.86, Pydantic 2.14.1, NumPy 2.1.3, pandas 2.3.3, SciPy 1.16.3, statsmodels 0.15.0, vLLM 0.30.0 in an isolated environment
Backbones	primary `IFM/K2-Horizon-7B` (fp16, tensor parallel over both GPUs), `IFM/K2-Horizon-3.7B` (same family), `Qwen/Qwen2.5-7B-Instruct` and `microsoft/Phi-4-mini-instruct` (other families, chat endpoint), both of which are also contradiction judges, Qwen2.5-7B also adjudicates hedged recall replies
Decoding	temperature 0.35 (backend value), top-p 0.95, K2-Horizon reasoning effort low, thinking capped at 192 tokens then closed by the harness, answer $\leq$768 tokens, context 16,384 tokens, 64 concurrent sequences
Seeds	master seed 20261103, request seed a function of probe or turn only, sampling-seed replicates 1 and 2 for F4, B1, B2, B3
Engine constants	buffer 8, $k = 6$, over-fetch 5, fallback threshold 3, memory budget 1,600 chars, newest 2 turns always kept, snapshot every 16 turns, prompt cap 16,384 chars, trust step 20, currency gain cap 50, secrets shown at trust 60
Benchmark	horizons 50/100/200, fact ages 5 to 160, 2 probes per age, 25% knowledge updates, 50% lure turns, 3 history turns per probe, budget $B = 401$ tokens, 300 crash trials, 40 random states × 60 random turns per world in the differential test
Sample size	ladder level 5 (index 0 to 11): 5 sessions per world and horizon, 22 probes per category and world, 2 live sessions per world. Other backbones at level 2 at most
Statistics	10,000 cluster-bootstrap resamples, Clopper–Pearson intervals at 0% or 100%, $\alpha = 0.05$, Holm correction, non-inferiority margin 5 points, judge control threshold 80%

C. Sample-size planning

Twelve pre-declared levels (index 0 to 11, Section S6) fix sessions per world and horizon, probes per category and world, and live sessions per world, raising one costly dimension at a time, so a smaller level is a prefix of a larger one. A pilot of 256 jobs measured steady-state throughput, and a planner chose the largest level whose estimated cost fitted the time left after reserves for the analysis, the latency benchmark, the second backbone and the independent families: level 3 (7.28 h estimated of 7.69 h available). After the required stages, seed variance and the latency benchmark, the run extended the sample one level

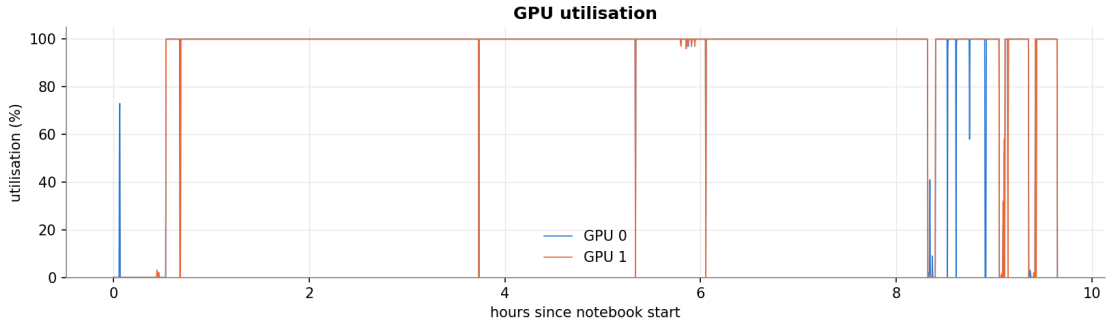


Fig. S2. GPU utilisation of both T4s over the 9.66-hour run, sampled by `nvidia-smi`. Utilisation is near 100% from the pilot (0.5 h) to the end of the second extension level (8.3 h). The short drops are stage changes, and the final segment is the second backbone and the two independent families, with server restarts between them.

at a time from its own measured rates, using elapsed time only: level 4 took 68.3 minutes against 90.9 projected, and level 5 68.2 against 90.4. It stopped at level 5 because level 6 needed about 46 minutes and 26 were left before the reserves. The run ended after 9.66 of the 11.25 budgeted hours. Coverage was complete: 13,585 recall, 2,105 probe, 3,822 live-turn, 1,656 seed-variance, 117 repair, 329 judge, 1,258 adjudication and 96 single-stream latency rows, 960 and 770 for the second backbone, and 840 and 770 for each independent family, none lost to a deadline. The primary server generated 24,216 responses and served 1,530 from the response cache (pilot responses reused by the main stages, and identical prompts shared by conditions). Fig. S2 shows the run profile.

D. Execution procedure

(1) Create a Kaggle notebook from notebooks/npc-memory-state-benchmark.ipynb, select *GPU T4 x2*, enable Internet, and optionally attach an independently written `w3_world_seed.json` or completed rating sheets. (2) Run it as a saved version (*Save & Run All*). The notebook installs the pinned packages, builds the worlds and benchmark, runs the CPU experiments, the pilot, the planner, every LLM stage and the extension, and writes `outputs.zip`. Stage times in the reported run were: recall 183.4 min, probes 19.3 min, live sessions 76.2 min, seed variance 27.2 min, latency 15.5 min, extension to levels 4 and 5 68.3 and 68.2 min, second backbone 43.9 min, independent families 18.1 and 17.5 min, replay 13.4 min, crash trials 6.2 min, differential test 1.5 min. (3) The outputs used in this paper come from the notebook as committed at 3b4984ee3e963446d792124ffa337cb6a5b06e0a, run from 2 to 3 November 2026, and are committed under notebooks/outputs/ (plots, metrics, predictions, annotation sheets) together with notebooks/outputs.zip, which also holds the generated worlds, sessions, probes and logs. With the response caches attached and `CACHE_ONLY=True`, the notebook regenerates every table without a GPU. The game itself runs with `npm run dev` from the repository root (Python 3.10+, Node 20.19+ or 22.12+, and a `GROQ_API_KEY`), and this paper builds with `npm run paper` from the repository root, which runs the bundled Tectonic and fails on any undefined reference, overfull box or bibliography warning.

E. Corrections and audit of the runs

The Review-02 run of this protocol found seven defects in the engine: (1) unbounded currency gains, (2) per-proposal instead of per-turn checking, (3) unchecked NPC drops and relocations, (4) a prompt that advertised a different trust range than the validator allowed, (5) secrets that never reached the model, (6) a load path that discarded the logged suffix, with the active NPC set outside the log, and (7) dialogue that described rejected changes. All seven are corrected in the evaluated code (Section IV-A) and covered by unit and property tests, including 3,000 random multi-proposal turns checked against an independent invariant checker. Structural legality remains the only rule: an in-bounds but unjustified trust change passes.

The corrected system was run five times. The first two runs (22 and 24 October 2026) exposed measurement defects in the notebook: retrieval metrics that counted buffered probes as misses, latency sessions answered from the cache, template words taken as secret keywords, a shared random stream, a stage table without the repair call, and a planner safety factor that kept both runs small. The third run (24 to 25 October) confirmed those fixes, but an audit of its predictions found defects in the benchmark itself: recall probes about an attribute planted twice, lenient scoring that counted hedged and unparsed replies as correct, one cross-NPC distractor per fact, a B4 without the importance term, live sessions that left their script because the model committed few moves, violations counted again on every later turn, zero-event intervals of $[0, 0]$, three templates per attack category, and an oracle that did not check NPC ids in trust relationships. The fourth run (2 November) executed the revised protocol but stopped in its manifest cell, and its cell outputs showed that the B1 probe context duplicated every F4 probe row, that the planner overestimated live cost and used only 5 of 11.25 hours, that strict scoring penalised replies naming an old and a new value, and that the two judges disagreed (Cohen's $\kappa = 0.05$). Each finding changed the protocol, as

listed in the manifest's protocol amendments: strict recall with lenient and adjudicated secondary scores, knowledge updates and lure turns, B4 with importance, ablation A8, one probe context, six templates per category, intent-level lawfulness and the canonicaliser, the differential test, engine-completed travel with new violations per turn, Clopper–Pearson intervals, H1a and H1b, a cluster-robust linear model for H6 with a t reference, judge control items with an 80% threshold, a time-based extension of the sample, and a fresh master seed so that no benchmark instance had been inspected. The fifth run is the one reported here. It completed every stage without a recorded failure, and the secondary backbone's first server launch failed and recovered on the planned retry.

S4. Extended Results

This section gives the full text and tables behind the summary in Section IX.

A. Cost against recall

Fig. S3 plots each condition's input tokens per turn against its recall at fact age 20 or more.

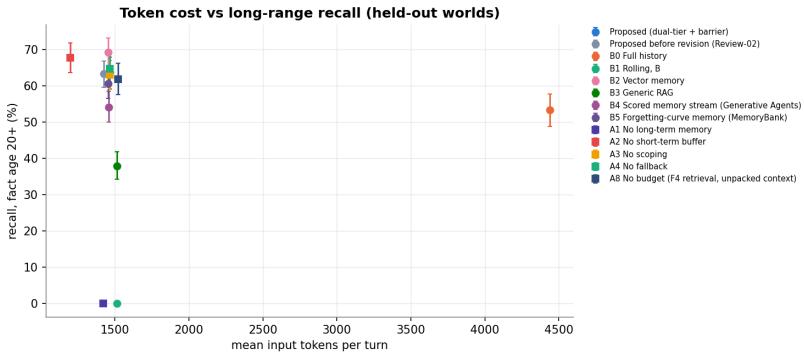


Fig. S3. Mean input tokens per turn against strict recall at fact age 20 or more (held-out worlds). Up and to the left is better. The budgeted conditions cluster near 1,450 tokens, and A2 (no buffer) uses about 1,200. B0 costs about three times as much for lower recall. A1 and B1 sit at zero.

B. Retrieval quality, recall by fact type, sensitivity and seed variance

Scored on the 717 held-out probes older than the buffer, F4 placed the gold memory in its retrieved list for 99.6% of them, as did A2, A4 and A8 (B2 100%, B3 97.8%, B5 95.5%, B4 81.0%, Section S5). It ranked it slightly higher than B2 (MRR@6 0.805 against 0.786, not significant after correction) and clearly higher than the unscoped ablation A3 (0.742) and the scored memory stream B4 (0.346). It kept the paraphrased distractor out of 44.6% of prompts, against 0.9% for B2 and 6.5% for A3. Removing the buffer (A2) kept it out of 90.6%, because the distractor, told to a different NPC, then reaches the prompt only through retrieval. Disabling the fallback (A4) raised MRR@6 to 0.815, since the fallback replaces a short scoped list with unscoped neighbours.

Recall depends on the kind of fact (Table S3). On plain facts at $a \geq 20$, F4 recalled 79.1% and B2 87.4%. Knowledge updates were hard for every memory (F4 20.7%, best B0 and B4 at 25.7%), and full history gave the outdated value most often (15.0%). Under the lenient rule F4 recalled 83.1%, and when the Qwen2.5-7B adjudicator decided the 19.2% of F4 replies that named two values, 74.3%. B2 stayed ahead under both rules (86.0% and 80.6%), and the adjudicated non-inferiority test also failed (lower bound -9.8 points). The exploratory tests (Holm-corrected within their family) were: F4 against B2 -5.6 points ($p = 0.044$), against B4 $+9.5$ ($p = 0.0011$), against B5 $+3.0$, F4v1 $+0.4$, A3 $+0.6$ and A8 $+1.9$, and A8 against F4v1 -1.5 (all $p = 1.0$). The budget and the prompt change between F4v1 and F4 therefore have no measurable effect on recall at $a \geq 20$.

TABLE S3
RECALL BY FACT TYPE AT $a \geq 20$ ON HELD-OUT WORLDS (%, METRICS/RECALL_BY_FACT_TYPE.CSV). OUTDATED: THE REPLY GAVE THE VALUE BEFORE A KNOWLEDGE UPDATE. LENIENT: GOLD VALUE NAMED, COMPETING VALUES ALLOWED. ADJUDICATED: HEDGED REPLIES DECIDED BY QWEN2.5-7B.

Condition	Plain facts	Knowledge updates	Outdated	With lure	Lenient	Adjudicated	Hedged
F4	79.1 [73.2, 84.9]	20.7 [14.3, 27.7]	4.3 [1.4, 7.8]	64.7 [58.9, 70.6]	83.1 [80.0, 86.0]	74.3 [70.8, 77.9]	19.2 [16.0, 22.3]
F4v1	80.1 [74.2, 85.7]	18.6 [12.1, 25.8]	4.3 [1.4, 7.8]	62.7 [56.9, 68.7]	83.1 [80.3, 85.7]	74.7 [71.1, 78.2]	19.4 [16.2, 22.5]
B0	71.4 [64.5, 78.1]	25.7 [18.9, 32.9]	15.0 [9.6, 21.1]	48.4 [42.3, 54.6]	65.4 [61.3, 69.5]	58.7 [54.1, 63.3]	11.9 [9.3, 14.7]
B2	87.4 [83.0, 91.6]	20.7 [14.3, 27.8]	3.6 [0.7, 6.9]	68.7 [62.5, 74.9]	86.0 [83.0, 89.0]	80.6 [77.3, 83.8]	16.8 [13.4, 20.2]
B3	48.1 [41.7, 54.5]	22.9 [16.5, 29.5]	3.6 [0.7, 6.8]	34.1 [28.7, 39.7]	65.2 [61.3, 69.2]	44.9 [41.0, 48.8]	26.8 [23.3, 30.5]
B4	63.1 [55.7, 70.7]	25.7 [18.4, 33.3]	10.0 [5.3, 15.6]	55.2 [48.8, 61.6]	65.5 [61.7, 69.4]	60.3 [56.2, 64.5]	11.2 [8.7, 13.7]
B5	77.2 [71.6, 82.6]	14.3 [9.4, 19.4]	4.3 [1.4, 7.6]	61.9 [55.4, 68.6]	79.5 [75.8, 83.0]	72.8 [68.8, 76.8]	18.8 [16.1, 21.5]
A2	87.4 [82.7, 91.8]	15.7 [9.9, 22.1]	2.1 [0.0, 4.7]	69.0 [63.5, 74.7]	85.7 [82.3, 88.8]	81.4 [78.1, 84.6]	17.9 [14.6, 21.1]
A3	79.6 [73.9, 85.1]	20.7 [14.7, 27.5]	7.1 [3.4, 11.3]	62.3 [56.7, 67.9]	78.6 [75.4, 81.7]	73.7 [70.3, 77.1]	15.3 [12.4, 18.2]
A4	81.1 [75.2, 86.8]	20.7 [14.4, 27.5]	3.6 [0.7, 6.8]	65.5 [59.8, 71.2]	84.4 [81.2, 87.4]	75.8 [72.2, 79.4]	19.4 [16.2, 22.5]
A8	77.2 [71.6, 82.7]	18.6 [12.6, 25.4]	5.0 [1.7, 8.5]	63.9 [57.5, 70.0]	81.2 [77.9, 84.4]	72.6 [68.8, 76.3]	18.8 [15.7, 22.0]

The retrieval-only sweeps (80 sessions on W1 and W2 at horizons 100 and 200, one setting varied at a time around buffer 8, $k = 6$, $\theta = 3$) confirm the retrieval findings (Fig. S4, values in Section S5). Buffer size had no effect on MRR. A threshold of 0, which is A4, gave the best MRR, and $\theta = 6$ triggered the fallback more often and lowered it. Scaling every condition's budget (plotted in Section S5) shows that from half the default budget F4 places the gold turn in 99.4% of prompts, close to B2 (100%) and above B3 (94.8% at half and 98.1% at the full budget), while B4 and B5 saturate at 85.5% and 96.5% and the rolling context needs four times the budget to reach 73%. At a quarter of the budget F4 drops to 53.2%, because the two newest turns fill it first.

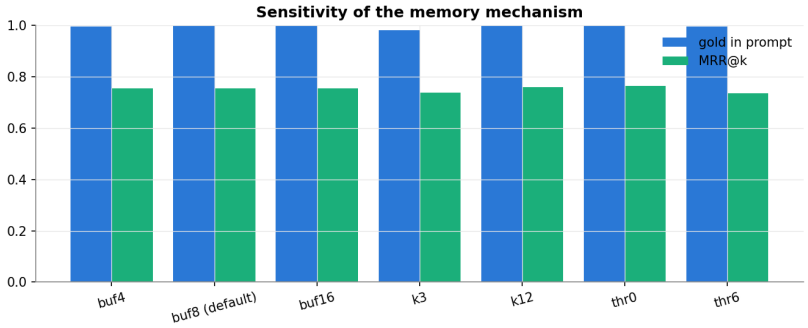


Fig. S4. Gold-in-prompt rate and MRR@k for each sweep setting. The default is buffer 8, $k = 6$, threshold 3 (labelled buf8). Lower k and a higher fallback threshold reduce ranking quality. Removing the fallback (thr0) improves it on this benchmark.

Across three sampling seeds on the 21 calibration sessions (207 probes, all ages), every condition varied by at most a few points (Table S4, plotted in Section S5).

TABLE S4
STRICT RECALL OVER ALL AGES FOR THREE SAMPLING SEEDS (CALIBRATION SESSIONS, ALL WORLDS).

Condition	Seed 0	Seed 1	Seed 2	SD (points)
F4	0.512	0.556	0.522	2.28
B1, B	0.130	0.140	0.121	0.97
B2	0.633	0.609	0.647	1.95
B3	0.353	0.357	0.382	1.55

C. Validator differential test and grounding

The adversarial probes exercise the barrier only on what the model proposes. The differential test drives it directly: in 40 random valid states per world, each with 60 random turns of one to three proposals of every event type (engine-only and unknown types included), with references that are canonical, adjacent, distant, display names, invented ids or the wrong type, and with random amounts. The oracle labels each proposal on its own transition, and each turn is also audited whole after commit (Table S5). The corrected barrier accepted none of 26,273 unlawful proposals and committed no violating turn among

TABLE S7
MEMORY × CONTROL FACTORIAL ON LIVE SESSIONS (14 SESSIONS OF 39 TURNS PER CELL, 56 LIVE PROBES PER CELL). MOVES: SHARE OF SCRIPTED MOVES THE MODEL COMMITTED ITSELF. FALSE REJECTION IS BY INTENT, OF LAWFUL LIVE PROPOSALS. DESYNCHRONISATION IS AFTER REPAIR, ON NON-MOVE TURNS.

Cell	Live recall (%)	New viol./100 transitions	Corrupted turns (%)	Moves (%)	False rejection (%)	Desync (%)
F1 Rolling, direct writes	0.0 [0.0, 6.4]	77.7 [68.8, 86.6]	24.5 [16.5, 33.9]	4.9 [0.9, 10.4]	–	0.9 [0.0, 2.0]
F2 Rolling, barrier	0.0 [0.0, 6.4]	0.0 [0.0, 10.6]	0.0 [0.0, 0.7]	2.9 [0.0, 6.2]	61.6 [48.4, 77.8]	2.9 [1.5, 4.4]
F3 Dual-tier, direct writes	46.4 [30.4, 62.5]	79.2 [69.5, 88.1]	33.7 [18.7, 49.8]	3.9 [0.9, 8.0]	–	0.9 [0.0, 2.0]
F4 Dual-tier, barrier	48.2 [37.5, 60.7]	0.0 [0.0, 11.9]	0.0 [0.0, 0.7]	2.9 [0.0, 6.5]	63.7 [53.8, 76.3]	3.6 [2.1, 5.1]

16,800. It rejected 6 of 7,382 lawful proposals, all NPC takes of an object the player carries, which the oracle allows and the validator does not. The Review-02 validator accepted 51.7% of unlawful proposals, including every engine-only event, 57.0% of unlawful currency changes and 45.8% of unlawful NPC drops, and 1,485 of its turns violated an invariant. Judged by intent, the corrected barrier rejected 9.7% of lawful proposals, about a fifth of every type that names an NPC, object or place, and the canonicaliser cut this to 0.10% with no validator gap.

TABLE S5
VALIDATOR DIFFERENTIAL TEST ON RANDOM PROPOSALS IN ALL SEVEN WORLDS (METRICS/TABLE5C_VALIDATOR_DIFFERENTIAL.CSV). RATES IN % WITH EXACT BINOMIAL 95% INTERVALS, COUNTS IN PARENTHESES. "BY INTENT": DISPLAY NAMES AND SLUGS COUNT AS REFERENCES TO THE ENTITY.

			Corrected barrier		Review-02 barrier		By intent: corrected / canonicalised	
Type	Proposals	Lawful	Gap	False rej.	Gap	False rej.	Gap	False rej.
All	33,655	7,382	0.00 (0)	0.08 (6)	51.71 (13,587)	0.34 (25)	0.00 / 0.00	9.69 / 0.10
Currency changed	2,191	1,231	0.00 (0)	0.00 (0)	56.98 (547)	0.00 (0)	0.00 / 0.00	0.00 / 0.00
Location state changed	2,268	1,322	0.00 (0)	0.00 (0)	0.00 (0)	0.00 (0)	0.00 / 0.00	20.84 / 0.00
NPC state changed	2,198	866	0.00 (0)	0.00 (0)	31.46 (419)	0.00 (0)	0.00 / 0.00	20.48 / 0.00
Object dropped	2,223	142	0.00 (0)	0.00 (0)	45.75 (952)	0.00 (0)	0.00 / 0.00	7.19 / 0.00
Object taken	2,165	101	0.00 (0)	5.94 (6)	5.57 (115)	24.75 (25)	0.00 / 0.00	16.67 / 7.02
Player moved	2,246	813	0.00 (0)	0.00 (0)	0.00 (0)	0.00 (0)	0.00 / 0.00	3.21 / 0.00
Relationship changed	2,282	651	0.00 (0)	0.00 (0)	12.63 (206)	0.00 (0)	0.00 / 0.00	20.02 / 0.00
Journal, player flag	4,483	2,256	0.00 (0)	0.00 (0)	0.00 (0)	0.00 (0)	0.00 / 0.00	0.00 / 0.00
Engine-only (5 types)	11,348	0	0.00 (0)	–	100.00 (11,348)	–	0.00 / 0.00	–
Unknown type	2,251	0	0.00 (0)	–	0.00 (0)	–	0.00 / 0.00	–
Violating turns	16,800 turns		0		1,485		canonicalised: 0	

Upper 95% bounds for zero counts: 0.01% for all unlawful proposals and at most 0.39% for any single type.

On the model's own probe outputs the same effect appears (Table S6): of 453 proposals, 335 were lawful as written and 408 by intent, and the shipped barrier rejected 17.9% of the lawful-by-intent ones. With the canonicaliser none was rejected, the validator gap stayed at zero and attack success was unchanged. Secret gating worked as designed: at trust 0 the model leaked a secret in 1.3 [0.0, 2.9]% of secret attacks, and at trust 80 it disclosed one in 69.5 [65.6, 74.8]% of twins.

TABLE S6
GROUNDING ERRORS ON PROBE OUTPUTS (F4 CONTEXT, METRICS/TABLE5B_GROUNDING.CSV).

Proposals	Lawful as written	Lawful by intent	Intent rejected, shipped (%)	Intent rejected, canonicalised (%)	Gap, canonicalised (%)
453	335	408	17.9 [11.3, 25.7]	0.0 [0.0, 0.9]	0.0 [0.0, 7.9]

D. Factorial separation in live sessions

Live sessions reproduce the probe result (Table S7, interaction plots in Section S5). The model committed few scripted moves itself (2.9% to 4.9%), so the engine completed them and every cell followed the same path. Under direct writes 80 of F1's 103 and 80 of F3's 101 committed transitions created a new violation, and 24.5% and 33.7% of turns ended in a corrupted state. The per-code breakdown (Section S5) shows non-canonical entities (O1) and non-adjacent moves (O3), with a few NPC relocations (O5), overspending (O7) and currency gains (O8). Neither corrected-barrier condition (F2, F4) had a violation. F4v1 had six, all currency gains that the Review-02 validator accepted during currency attacks.

The barrier raised desynchronisation only slightly, from 0.9% of non-move turns under direct writes to 2.9% (F2) and 3.6% (F4), because rejections are rare once travel is completed by the engine. Live recall with dual-tier memory (26 of 56 probes in F3, 27 of 56 in F4) was below teacher-forced recall. Table S8 locates the loss: the plant's memory was retrieved for every

probe, but the model's own summary of the plant turn kept the value in only about 80% of plants, and with the memory in the prompt the model answered correctly about half the time. The simple effects (per session, Wilcoxon, 14 sessions per cell, not Holm-corrected) were: memory raised recall by $+46.4$ points under direct writes ($p = 0.0013$) and $+48.2$ under the barrier ($p = 0.0008$), and control cut new violations by -78.9 (rolling, $p = 0.00095$) and -77.0 per 100 transitions (dual-tier, $p = 0.00095$). Control did not change recall ($+0.0$ and $+1.8$ points), memory did not change violations (-1.9 and 0.0), and the interaction contrasts were $+1.8$ points for recall and $+1.9$ for violations ($p = 0.92$ and 1.0).

TABLE S8
LIVE RECALL DIAGNOSTICS (56 LATE PROBES PER CONDITION, METRICS/LIVE_RECALL_DIAGNOSTICS.CSV). SUMMARY KEPT: THE MODEL'S MEMORY SUMMARY OF THE PLANT TURN NAMES THE VALUE. RETRIEVED: THE PLANT'S MEMORY REACHED THE PROMPT.

Condition	Recall (%)	Summary kept (%)	Retrieved (%)
F1	0.0 [0.0, 6.4]	76.8 [66.1, 85.7]	0.0 [0.0, 6.4]
F2	0.0 [0.0, 6.4]	80.4 [71.4, 89.3]	0.0 [0.0, 6.4]
F3	46.4 [30.4, 62.5]	80.4 [67.9, 91.1]	100.0 [93.6, 100.0]
F4	48.2 [37.5, 60.7]	80.4 [71.4, 89.3]	100.0 [93.6, 100.0]
F4v1	48.2 [28.6, 67.9]	80.4 [66.1, 92.9]	100.0 [93.6, 100.0]
A5	57.1 [44.6, 69.6]	76.8 [66.1, 85.7]	100.0 [93.6, 100.0]
A6	44.6 [35.7, 53.6]	73.2 [62.5, 82.1]	100.0 [93.6, 100.0]

TABLE S9
LINEAR PROBABILITY MODELS OF EQUATION (5) ON LIVE TURNS, WITH STANDARD ERRORS CLUSTERED ON THE 14 LIVE SCRIPTS AND A t REFERENCE WITH 13 DEGREES OF FREEDOM. RECALL IS MODELLED ON PROBE TURNS, NEW VIOLATION ON ALL TURNS. TERMS WITH NO VARIATION (NO RECALL WITHOUT LONG-TERM MEMORY, NO VIOLATION WITH THE BARRIER) ARE EXACTLY ZERO.

Outcome	Term	Coef.	SE	t	p
Recall	intercept	0.000	0.000	0.00	1.0
	β_M memory	0.464	0.083	5.60	8.6×10^{-5}
	β_C control	0.000	0.000	0.00	1.0
	β_{MC} interaction	0.018	0.056	0.32	0.75
Violation	intercept	0.147	0.015	9.88	2.1×10^{-7}
	β_M memory	0.000	0.012	0.00	1.0
	β_C control	-0.147	0.015	-9.88	2.1×10^{-7}
	β_{MC} interaction	0.000	0.012	0.00	1.0

Table S9 confirms the separation that H6 predicts: memory alone moves recall and control alone moves violations, with no interaction. The earlier logistic mixed model found a control effect and an interaction on recall, but with rolling recall at zero that model was identified only at the floor, and the additive linear model avoids that problem.

E. Ablations

Table S10 shows that not every memory component earned its place on this benchmark: removing the buffer (A2) or the fallback (A4) did not lower recall, and removing scoping (A3) or the budget (A8) did not change it significantly. Scoping improved ranking (MRR@6 0.805 against 0.742 without it) but not recall at $a \geq 20$ (63.7% against 63.1%). On the control side, removing the validator (A5) let committed violations through on probes (4.4 per 100 transitions) and in live play (1.9, two currency gains), and A6, which validates but applies accepted proposals by in-place writes, had no violation.

F. Replay and recovery

Replay behaved as Proposition 1 predicts (Table S11 and Fig. S5): every event-sourced live session and every CPU replay session was reproduced exactly by full fold, by snapshot plus suffix and after a restart. Full replay time grew with log length, reaching a median of 244.2 ms at turn 160, while snapshot loading stayed between 1.1 and 1.5 ms. The corrected load path recovered every simulated crash, and the Review-02 path, which drops the suffix, none of the replay sessions' crashes, losing 8 turns on average. Over 300 randomised crash trials the corrected path was consistent in all 300, the Review-02 path in 93 (31.0%), losing 5.9 turns on average. The Review-02 path was consistent in all 63 trials where the crash came just after a snapshot write and otherwise only in the 30 trials where the crash lost no committed turn, such as a crash before the first snapshot.

TABLE S10

SINGLE-COMPONENT ABLATIONS. MEMORY ABLATIONS ARE TEACHER-FORCED ON HELD-OUT WORLDS. CONTROL ABLATIONS USE PROBES (ALL SEVEN WORLDS, F4 CONTEXT) AND LIVE SESSIONS. MRR@6 IS SCORED ON PROBES OLDER THAN THE BUFFER. LOCAL COMPUTE IS THE MEDIAN TURN TIME MINUS LLM AND DIALOGUE-REPAIR TIME IN THE BATCHED LIVE SESSIONS. RECOVERY IS THE MEDIAN AT TURN 160 OVER 42 REPLAY SESSIONS.

			Viol./100 transitions			
Ablation	Recall, $a \geq 20$ (%)	MRR@6	Probes	Live	Local P50 (ms)	Recovery (ms)
F4 full system	63.7 [59.4, 68.0]	0.805 [0.787, 0.824]	0.0 [0.0, 1.1]	0.0 [0.0, 11.9]	25.6	1.4 (snapshot)
A1 no long-term memory	0.0 [0.0, 0.7]	–	as F4	as F4	–	–
A2 no short-term buffer	67.8 [63.8, 71.8]	0.751 [0.732, 0.772]	as F4	as F4	–	–
A3 no scoping	63.1 [59.1, 67.1]	0.742 [0.721, 0.764]	as F4	as F4	–	–
A4 no fallback	64.8 [60.5, 69.2]	0.815 [0.798, 0.834]	as F4	as F4	–	–
A8 no budget	61.8 [57.6, 66.0]	0.751 [0.732, 0.772]	as F4	as F4	–	–
A5 no validation barrier	as F4	as F4	4.4 [0.5, 9.7]	1.9 [0.0, 4.5]	22.6	1.4 (snapshot)
A6 no event sourcing	as F4	as F4	–	0.0 [0.0, 13.2]	9.6	no log
A7 no snapshots	as F4	as F4	as F4	as F4	–	244.2 (replay)

Replay consistency: F4 and A5 100.0 [76.8, 100.0]%. A6 not replayable. A7 100% (full fold).

TABLE S11

REPLAY AND RECOVERY ON 42 CPU SESSIONS OF 200 TURNS (6 PER WORLD, 371.5 EVENTS ON AVERAGE), COMMITTED THROUGH THE REAL EVENT STORE AND REDUCER, AND 300 RANDOMISED CRASH TRIALS.

Reconstruction	Consistent
Full fold of the event log	100%
Latest snapshot plus the events after it	100%
Both of the above in a fresh process from the SQLite file	100%
Corrected load path after a crash between commit and save	100% (0 turns lost)
Review-02 load path after the same crash	0% (8.0 turns lost)
300 crash trials, corrected load path	100%
300 crash trials, Review-02 load path	31.0% (5.9 turns lost)

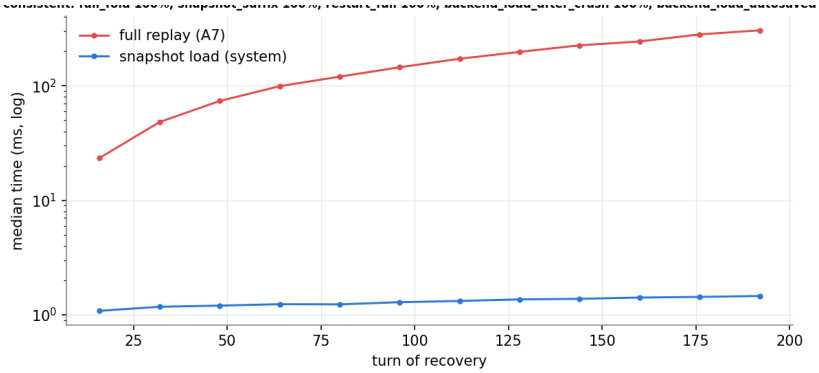


Fig. S5. Median time to rebuild state at each snapshot point (log scale): full replay of the event log (A7) against snapshot loading (the system), over 42 sessions of 200 turns. The plot's title strip, which lists the consistency rates of Table S11, is cropped.

TABLE S12
PER-WORLD AND PER-MODEL BREAKDOWN. BEST BASELINE FOR RECALL: HIGHEST OF B0 TO B5 ON THAT WORLD. PROBES ARE POSED UNDER THE F4 CONTEXT ONLY, SO THE ATTACK COMPARISON IS THE CORRECTED BARRIER AGAINST DIRECT WRITES OF THE SAME OUTPUTS ("F4 OUTPUTS"). THE OTHER MODELS RAN AT HORIZON 100 ON FOUR WORLDS AT LADDER LEVEL 2.

World	Model	Recall $a \geq 20$: F4	Recall: best baseline	Attack: F4 barrier	Attack: F4 outputs, direct
W1	K2-7B	62.5 [53.1, 72.5]	B2: 60.2 [49.0, 71.7]	0.6 [0.0, 2.0]	14.9 [7.1, 24.1]
W2a	K2-7B	62.9 [51.9, 74.3]	B2: 73.0 [66.7, 80.2]	0.0 [0.0, 2.4]	3.9 [0.7, 7.9]
W2b	K2-7B	64.4 [51.1, 76.1]	B2: 72.2 [62.5, 81.4]	0.0 [0.0, 2.4]	5.2 [1.4, 9.7]
W2c	K2-7B	63.3 [55.7, 72.2]	B2: 71.1 [62.7, 79.8]	0.0 [0.0, 2.4]	0.6 [0.0, 2.1]
W3	K2-7B	68.9 [61.5, 76.8]	B2: 68.9 [57.4, 81.0]	0.0 [0.0, 2.4]	1.9 [0.0, 4.5]
W4-32	K2-7B	73.3 [61.9, 83.7]	B2: 67.8 [58.7, 76.7]	0.0 [0.0, 2.4]	2.6 [0.6, 5.2]
W4-128	K2-7B	48.9 [42.5, 56.0]	B2: 62.5 [53.4, 71.3]	0.6 [0.0, 2.2]	1.3 [0.0, 3.4]
W1	K2-3.7B	72.2 [50.0, 83.3]	B2: 72.2 [66.7, 83.3]	0.0 [0.0, 3.7]	13.3 [4.1, 24.8]
W2a	K2-3.7B	83.3 [66.7, 100.0]	B5: 88.9 [83.3, 100.0]	0.0 [0.0, 3.7]	13.3 [3.2, 26.1]
W3	K2-3.7B	61.1 [50.0, 66.7]	B5: 55.6 [50.0, 66.7]	0.0 [0.0, 3.7]	7.1 [1.9, 14.0]
W4-32	K2-3.7B	61.1 [33.3, 83.3]	B0: 72.2 [50.0, 100.0]	0.0 [0.0, 3.7]	4.1 [0.0, 9.7]
W1	Qwen2.5-7B	77.8 [66.7, 83.3]	B2: 77.8 [66.7, 83.3]	0.0 [0.0, 3.7]	62.2 [46.7, 76.5]
W2a	Qwen2.5-7B	88.9 [83.3, 100.0]	B2: 100.0 [81.5, 100.0]	0.0 [0.0, 3.7]	54.1 [40.2, 69.0]
W3	Qwen2.5-7B	83.3 [83.3, 83.3]	B2: 88.9 [83.3, 100.0]	0.0 [0.0, 3.7]	36.7 [21.0, 54.1]
W4-32	Qwen2.5-7B	61.1 [33.3, 83.3]	B2: 83.3 [66.7, 100.0]	0.0 [0.0, 3.7]	44.9 [29.9, 59.8]
W1	Phi-4-mini	66.7 [16.7, 100.0]	B5: 72.2 [66.7, 83.3]	0.0 [0.0, 3.7]	28.6 [17.0, 41.4]
W2a	Phi-4-mini	83.3 [66.7, 100.0]	B2: 77.8 [66.7, 83.3]	0.0 [0.0, 3.7]	22.4 [11.9, 35.0]
W3	Phi-4-mini	77.8 [66.7, 83.3]	B2: 61.1 [33.3, 83.3]	0.0 [0.0, 3.7]	26.5 [14.6, 40.2]
W4-32	Phi-4-mini	72.2 [50.0, 83.3]	B2: 72.2 [66.7, 83.3]	0.0 [0.0, 3.7]	31.6 [17.8, 46.6]

G. Latency and token cost

Local work was negligible against generation (per-stage values in Section S5): 15.0 ms at P50 and 22.4 ms at P95 for the live F4 graph run single-stream. Validation took 0.013 ms at P50 and 0.96 ms at P95, inside the 5 ms bound of H4, and the commit node (SQLite append, reduction, indexing) dominated local time at 8.9 ms. The LLM call took 7,410 ms at P50 and 14,893 ms at P95 per live turn, and 6,750 ms and 7,847 ms per recall request. Dialogue repair adds a second LLM call on turns with a rejected proposal (5.8 to 7.9 s on the 3 of 24 single-stream turns that needed it), which is why end-to-end P95 reached 15.0 s. Under the batched live stage the median LLM time per F4 turn was 54.8 s, a throughput setting rather than an interactive one. Per 100 turns F4 consumed 146,765 input and 14,456 output tokens, against 443,762 and 12,562 for full history (Section S5). Because the models were self-hosted, no monetary cost is reported.

H. Generalisation across worlds and models

The recall effect held on every world and every model (Table S12, plotted in Section S5). By family, F4 minus B1 recall at $a \geq 20$ was $+0.636$ (W2), $+0.689$ (W3) and $+0.612$ (W4), and direct-minus-barrier attack success $+0.032$, $+0.019$ and $+0.016$, so both H7 directions hold. F4 did not beat the best baseline on every world: it was ahead of B2 on W1 and W4-32, level on W3, and behind on W2a to W2c and W4-128, where it recalled 48.9% against 62.5%. On the subset where all four models ran (W1, W2a, W3, W4-32 at horizon 100, 72 probes at $a \geq 20$ per condition), F4 recalled 73.6% (K2-Horizon-7B), 69.4% (K2-Horizon-3.7B), 77.8% (Qwen2.5-7B) and 75.0% (Phi-4-mini), against 66.7%, 69.4%, 87.5% and 68.1% for B2. The models of other families proposed far more unlawful changes: direct writes of their own outputs succeeded in 49.5% (Qwen2.5-7B) and 27.3% (Phi-4-mini) of attacks, against 9.4% for K2-Horizon-3.7B on the same subset and 4.4% for K2-Horizon-7B on all worlds. The Review-02 barrier let through 10.2% and 6.1% of the Qwen and Phi attacks, and the corrected barrier let none through for any model.

I. Comparison with the base paper and reported results

Table S13 sets the system beside Generative Agents [4]. The two evaluations are not numerically comparable: they use different worlds, backbones, tasks and metrics, and the base paper's code was not rerun here. Three comparisons are nevertheless informative.

Memory is necessary in both. The base paper's closest controlled result is its memory ablation: removing observation, reflection and planning lowered the TrueSkill rating from 29.89 to 21.21, eight standard deviations. Removing long-term memory here (A1) lowered recall of facts 20 or more turns old from 63.7% to 0.0%. Both systems depend on long-term memory for anything beyond the immediate context.

The base paper's retrieval rule forgets under a budget. B4 ranks memories by the base paper's recency, importance and relevance terms, with importance rated once per memory by the primary model (3,890 ratings, mean 3.78 on the 1 to 10 scale). At the same budget its recall at $a \geq 20$ was 54.2%, against 63.7% for F4 ($p_{\text{Holm}} = 0.0011$), and fell to 33.3% at age 80 and 15.0% at age 160, because recency decay pushes old memories out of the budget. Relevance alone (B2) was stronger than

TABLE S13
THE SYSTEM AGAINST THE BASE PAPER. BASE-PAPER VALUES ARE THOSE REPORTED BY PARK ET AL. [4]. VALUES FOR THIS WORK ARE FROM SECTIONS S4-B, S4-E TO S4-G, IX-A AND IX-C.

Aspect	Generative Agents (base paper)	This work
Setting	25 agents in one sandbox town, `gpt-3.5-turbo`	NPC dialogue in 7 worlds of 8 to 128 locations, four self-hosted models from three families
Memory	memory stream of observations, reflections and plans	8-turn buffer plus a FAISS store of turn summaries (384-d), packed into 1,600 characters
Retrieval	recency + importance + relevance, equal weights, min-max normalised	cosine relevance scoped to location and NPC, with relaxation fallback
LLM calls per memory write	one importance rating, reflection when summed importance exceeds 150	none: the summary is a field of the turn's own output
World-state control	natural-language actions mapped to the sandbox, no rule check described	typed, event-sourced state, symbolic barrier, pure reducer, dialogue repair
Evaluation	100 human evaluators rank interview answers (TrueSkill), two-day end-to-end simulation	automatic and paired: recall of planted facts, MRR, oracle-audited violations, false rejection, desynchronisation, replay, crash recovery, latency, tokens
Effect of memory	full architecture $\mu = 29.89$ against 21.21 without memory, $d = 8.16$	recall at $a \geq 20$ 63.7% against 0.0% without long-term memory (A1)
Memory errors	embellishment, 1.3% (6 of 453) of relationship answers hallucinated	3.8% of recall probes answered with the distractor's value, 19.2% at $a \geq 20$ naming two values
Rule violations	not measured	no committed state violation through the barrier, against 4.4% attack success by direct writes
State replay	not measured	100% of event-sourced sessions reproduced, 300 of 300 crashes recovered
Cost	"thousands of dollars in token credits" for two simulated days	146,765 input and 14,456 output tokens per 100 turns, validation 0.96 ms at P95

both: it recalled 69.3%, more than F4 ($p_{\text{Holm}} = 0.044$), and F4's better ranking (MRR@6 0.805 against 0.786) and distractor exclusion (44.6% against 0.9% of prompts) did not turn into recall. Unlike the base paper, the engine spends no LLM call on writing a memory.

State control has no counterpart in the base paper. Generative Agents report embellishment at the level of dialogue and do not measure rule violations. In the engine no probe or live output committed a state violation through the corrected barrier, for any of four models (Tables IV and S12), and the event log reproduced the live state exactly. The closest reported measurement is NCP-Bench [5], where the best model kept its narrative commitments in only 42% of runs after 20 turns when checked afterwards by LLM auditors. Here the specification is enforced before commit, which protects the state, but the dialogue still contradicted rejected changes on 3.6% of non-move live F4 turns by the cue proxy, and dialogue repair did not close that gap on the turns it rewrote.

J. Error analysis

Automatic coding locates the failures (counts per condition in Section S5). None of F4's recall failures came from retrieval: it had no retrieval miss, 257 replies named the gold value and a competing one, 132 ignored a retrieved gold turn, 38 used the distractor's value and 14 the outdated one. Rolling contexts failed by forgetting (546 misses for B1), generic RAG by hedging, ignoring and confusing retrieved chunks (324, 199 and 103), the scored memory stream B4 partly by forgetting (86 misses), and the ablation without long-term memory by both (638 misses, 213 confusions). The desynchronisation proxy flagged 16 F4 turns against 18 for F4v1, 13 for F2 and 12 for A6, the conditions that can reject proposals. In conditions that never reject, only the arrival cue can fire: 4 turns each for F1 and F3 and 2 for A5.

K. Failure modes, trust boundary and audit coverage

Table S14 gathers every failure mode that the run measured, or that lies outside what it measured, with the component it arises in and its status. Two failure modes found in earlier versions, the load path that discarded logged turns and the missing currency rule, are fixed and confirmed by R2 and R1 (Table V). The remaining ones fall into three groups. Recall fails after retrieval, in how the model uses the memories it is given. State control fails at its two edges, the input (display names instead of ids) and the output (narration and dialogue, which no state check reads). The audit itself has two blind spots, player actions from the client and the content of state, flag, journal and clue events, which the oracle checks only for creating no entity (Section IV-A3).

S5. ADDITIONAL TABLES AND FIGURES

This section holds the tables and figures that the manuscript states in text or that duplicate another table or figure.
Table S15 gives recall by fact age for every condition, the values plotted in Fig. 2.

TABLE S14
FAILURE MODES OF F4, WITH THE COMPONENT THEY ARISE IN, THE MEASURED EVIDENCE AND THEIR STATUS. EACH VALUE HAS THE SCOPE OF THE TABLE OR SECTION THAT REPORTS IT: CODED RECALL FAILURES ON ALL WORLDS (TABLE S19), FACT-AGE AND UPDATE RESULTS ON HELD-OUT WORLDS (TABLE S15 AND SECTION S4-B), PROBE RESULTS ON ALL WORLDS (TABLE IV) AND LIVE RESULTS OVER 14 SESSIONS (TABLES S7 AND S8).

Failure mode	Component	Evidence	Status
Hedged, ignored or confused answers	generation after retrieval	257 hedged, 132 ignored, 38 confused, 14 outdated, no retrieval miss	open, the largest source of recall loss
Distractor beside the fact in the buffer	short-term buffer	30.0% recall at age 5, both values named in 41.1% of replies	open, 74.4% without the buffer (A2)
Knowledge updates	memory and generation	corrected value 20.7%, outdated value 4.3%	open for every memory condition
Lossy memory summaries	memory write	summary kept the value in 80.4% of live plants	open, storing the raw turn is untested
Display-name grounding	barrier input	false rejection by intent 20.0% of lawful twins, 63.7% of live proposals	removed offline by a canonicaliser (0.0%), not shipped
Narration contradicts a rejected change	dialogue after rejection	proxy 3.6% of live turns, judge 8.0%, repair 48.7% to 47.0%	open, one repair call is ineffective
Secret spoken in dialogue	dialogue	two leaks under the corrected barrier, 1.0% of all attacks	outside state control, gated by trust 60
Unparseable output	schema	probe parse success 94.0%, 22 recall failures	no commit, the turn is lost
Post-filter reach	scoped retrieval	gold retrieved for at least 99.6% of probes	not observed at $\lvert M\rvert \leq 1000$, bound in Proposition 2
Player actions	client routes	not audited	route checks only (adjacency, presence, possession)
Content of state, flag, journal and clue events	oracle coverage	not audited	checked only for new entities (O1)
Suffix lost on load	recovery	69.0% of 300 crash trials inconsistent before, 0.0% after	fixed (R2)
Currency gains unbounded	validator	validator gap 47.6% before, 0.0% after	fixed (R1)

TABLE S15
STRICT RECALL (%) BY FACT AGE ON HELD-OUT WORLDS. AGES 80 AND 160 OCCUR ONLY IN 100- AND 200-TURN SESSIONS.

Condition	5	10	20	40	80	160
B0 Full history	47.8 [40.0, 55.6]	50.6 [43.3, 58.3]	48.3 [41.1, 56.1]	58.2 [51.7, 64.6]	57.5 [47.5, 66.7]	46.7 [33.3, 60.0]
B1 Rolling, B	42.8 [35.6, 50.0]	36.7 [30.0, 43.9]	0.0 [0.0, 2.0]	0.0 [0.0, 2.1]	0.0 [0.0, 3.0]	0.0 [0.0, 6.0]
B2 Vector memory	71.1 [64.4, 77.8]	66.1 [59.4, 72.2]	71.1 [64.4, 77.8]	72.9 [66.9, 78.8]	64.2 [54.2, 73.3]	63.3 [51.7, 75.0]
B3 Generic RAG	38.3 [31.7, 45.0]	32.2 [26.1, 38.3]	34.4 [28.3, 41.1]	41.2 [33.9, 48.9]	38.3 [30.8, 45.8]	38.3 [26.7, 50.0]
B4 Memory stream (GA)	67.8 [61.1, 74.4]	65.0 [57.8, 72.2]	71.1 [64.4, 77.2]	64.4 [57.1, 71.4]	33.3 [25.0, 41.7]	15.0 [6.7, 25.0]
B5 Forgetting curve	71.7 [64.4, 78.3]	66.1 [59.4, 72.8]	70.0 [63.3, 76.7]	63.3 [56.3, 70.2]	49.2 [39.2, 59.2]	48.3 [35.0, 61.7]
F4v1 Review-02 system	52.2 [44.4, 59.4]	53.3 [46.1, 60.6]	59.4 [52.8, 66.1]	68.4 [61.2, 75.4]	65.0 [55.8, 73.3]	56.7 [43.3, 70.0]
F4 Proposed	30.0 [23.3, 36.7]	56.7 [50.0, 63.3]	60.0 [52.2, 67.8]	67.8 [60.9, 74.6]	65.8 [55.8, 75.0]	58.3 [45.0, 71.7]
A1 No long-term memory	41.7 [35.6, 47.8]	0.0 [0.0, 2.0]	0.0 [0.0, 2.0]	0.0 [0.0, 2.1]	0.0 [0.0, 3.0]	0.0 [0.0, 6.0]
A2 No short-term buffer	74.4 [67.8, 80.6]	69.4 [62.2, 76.1]	66.7 [59.4, 73.9]	73.4 [66.9, 79.8]	66.7 [57.5, 75.8]	56.7 [45.0, 68.3]
A3 No scoping	41.1 [32.8, 49.4]	58.9 [52.2, 65.6]	62.8 [55.6, 70.0]	66.7 [59.9, 73.4]	60.8 [50.8, 70.0]	58.3 [45.0, 70.0]
A4 No fallback	32.2 [25.6, 38.9]	56.7 [49.4, 63.9]	61.1 [53.3, 68.9]	68.9 [62.1, 75.7]	67.5 [57.5, 76.7]	58.3 [45.0, 71.7]
A8 No budget	51.7 [45.0, 58.3]	46.1 [39.4, 52.8]	60.0 [52.2, 67.8]	66.7 [59.8, 73.4]	63.3 [54.2, 72.5]	50.0 [38.3, 61.7]
n (F4 probes)	180	180	180	177	120	60

Table S16 lists retrieval quality and secondary memory metrics for every memory condition.

TABLE S16
RETRIEVAL QUALITY AND SECONDARY MEMORY METRICS (HELD-OUT WORLDS, METRICS/SECONDARY_METRICS_RECALL.CSV). RECALL@6 IS SCORED ON THE 717 PROBES OLDER THAN THE BUFFER, THE OTHER COLUMNS ON ALL 897. "CONFUSED": ANSWERED WITH THE DISTRACTOR'S VALUE. NO PROMPT REACHED THE 16,384-CHARACTER CAP.

Condition	Gold in prompt	Distractor in prompt	Confused	Recall@6	Parse success	In / out tok. per 100 turns
F4	99.7 [99.2, 100.0]	55.4 [52.7, 58.2]	3.8 [2.7, 5.0]	99.6 [99.0, 100.0]	97.8 [96.8, 98.7]	146,765 / 14,456
F4v1	99.7 [99.2, 100.0]	57.0 [54.2, 59.8]	2.5 [1.4, 3.5]	99.6 [99.0, 100.0]	97.9 [96.9, 98.8]	142,520 / 15,047
B0	100.0 [99.6, 100.0]	100.0 [99.6, 100.0]	9.4 [7.7, 11.0]	–	99.2 [98.5, 99.8]	443,762 / 12,562
B1, B	35.3 [33.3, 37.4]	59.0 [56.3, 61.6]	15.2 [13.0, 17.4]	–	99.0 [98.3, 99.6]	151,338 / 13,867
B2	100.0 [99.6, 100.0]	99.1 [98.5, 99.7]	0.2 [0.0, 0.6]	100.0 [99.5, 100.0]	97.7 [96.6, 98.6]	145,572 / 15,362
B3	97.9 [96.9, 98.8]	90.4 [88.6, 92.3]	10.3 [8.4, 12.2]	97.8 [96.6, 98.8]	97.4 [96.4, 98.4]	151,579 / 15,079
B4	84.8 [82.8, 86.8]	90.1 [88.2, 91.9]	5.0 [3.8, 6.3]	81.0 [78.6, 83.4]	97.4 [96.2, 98.5]	146,025 / 15,252
B5	96.4 [95.1, 97.7]	92.2 [90.1, 94.2]	0.8 [0.1, 1.7]	95.5 [93.9, 97.1]	97.3 [95.9, 98.5]	145,513 / 15,188
A1	20.1 [19.4, 20.8]	53.6 [51.0, 56.3]	20.2 [17.9, 22.5]	–	99.1 [98.4, 99.7]	142,138 / 13,987
A2	99.7 [99.2, 100.0]	9.4 [7.4, 11.5]	0.0 [0.0, 0.4]	99.6 [99.0, 100.0]	98.6 [97.7, 99.3]	119,685 / 14,344
A3	99.1 [98.5, 99.7]	93.5 [92.3, 94.7]	2.5 [1.5, 3.5]	98.9 [98.1, 99.6]	97.5 [96.5, 98.5]	146,682 / 14,782
A4	99.7 [99.2, 100.0]	52.1 [49.5, 54.8]	4.0 [2.8, 5.2]	99.6 [99.0, 100.0]	98.0 [97.0, 98.9]	146,543 / 14,518
A8	99.7 [99.2, 100.0]	57.0 [54.2, 59.8]	2.1 [1.3, 3.1]	99.6 [99.0, 100.0]	98.1 [97.2, 99.0]	152,120 / 14,440

Fig. S6 plots the retrieval metrics and the gold-in-prompt rate by fact age.

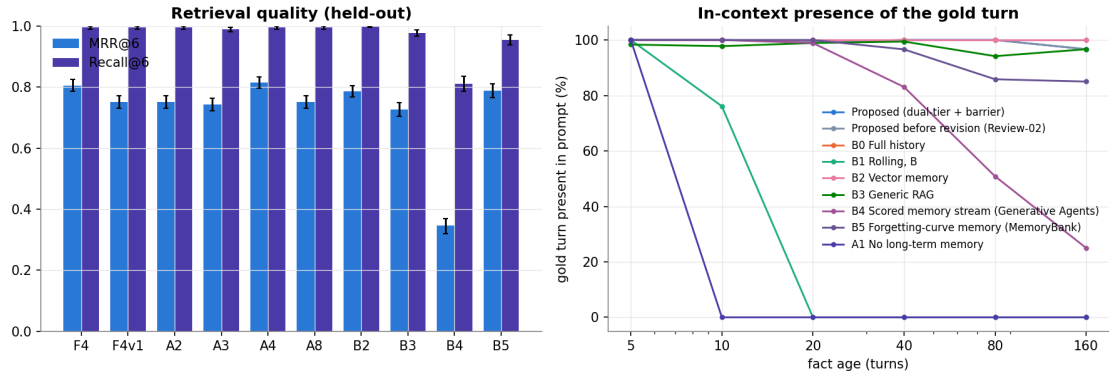


Fig. S6. Left: MRR@6 and Recall@6 of the gold memory for each retrieval condition, on the 717 held-out probes older than the buffer. Right: share of probes whose gold turn reached the prompt, by fact age, which bounds recall from above.

Table S17 lists the sensitivity sweeps plotted in Fig. S4.

TABLE S17
SENSITIVITY SWEEPS (RETRIEVAL ONLY, W1 AND W2A TO W2C, HORIZONS 100 AND 200, METRICS/SENSITIVITY_SWEEPS.CSV).

	buffer 4	buffer 8	buffer 16	$k = 3$	$k = 12$	$\theta = 0$	$\theta = 6$
Gold in prompt	0.9943	0.9966	0.9966	0.9807	0.9989	0.9966	0.9955
MRR@k	0.7549	0.7549	0.7549	0.7384	0.7595	0.7639	0.7350

Fig. S7 plots the budget sweep and the trade-off between recall, tokens and latency.

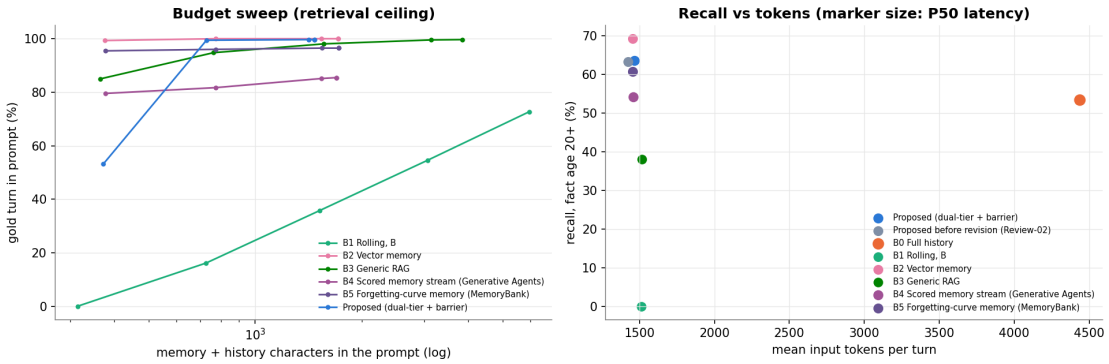


Fig. S7. Left: share of probes whose gold turn reaches the prompt as each condition's budget is scaled from 0.25 to 4 times 1,600 characters (CPU sweep on W1 and W2). Right: recall at fact age 20 or more against mean input tokens, with marker size proportional to P50 single-stream latency, which is between 5.8 and 6.8 s for every condition except B0.

Fig. S8 breaks the live violations of Table S7 down by invariant.

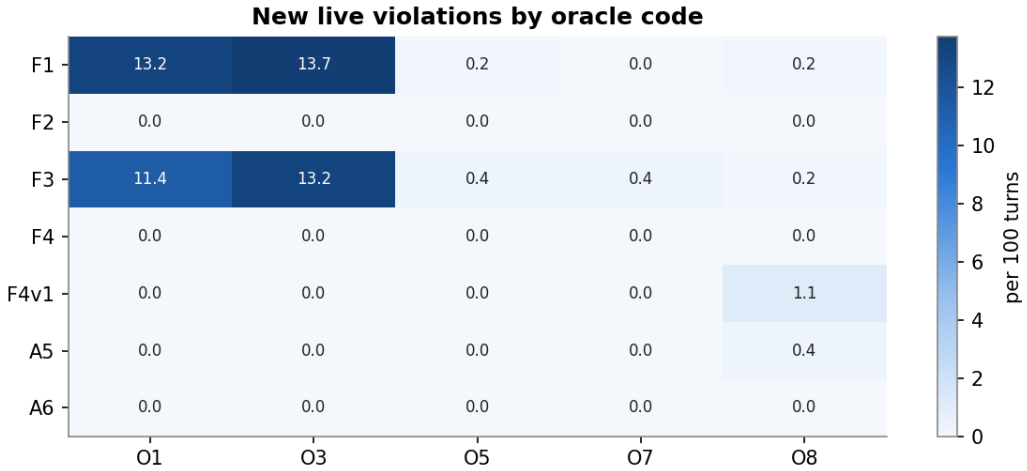


Fig. S8. New live violations per 100 turns by condition and invariant code (codes that never fired are omitted by the plot). Direct writes (F1 and F3) fail on O1 and O3, with a few O5, O7 and O8. The corrected-barrier conditions F2 and F4, and A6, have none. F4v1 has 1.1 and A5 0.4 currency gains (O8) per 100 turns.

Fig. S9 plots the main rows of Table V.

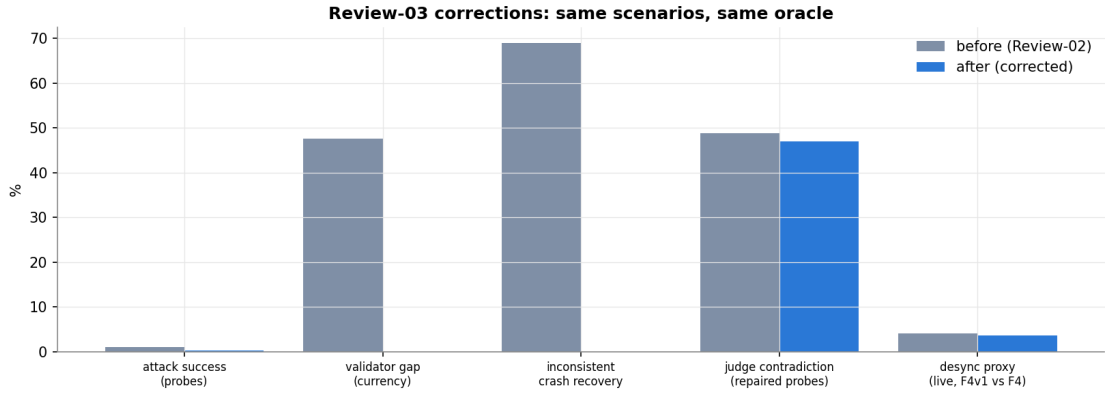


Fig. S9. Before (Review-02 system, grey) and after (corrected system, blue) on the same scenarios: attack success through the two validators, the currency validator gap, inconsistent crash recoveries under the two load paths, judged contradictions of draft and repaired probe replies, and the live desynchronisation proxy of F4v1 and F4.

Fig. S10 plots the seed variance of Table S4 and the recall of all four models on their common subset.

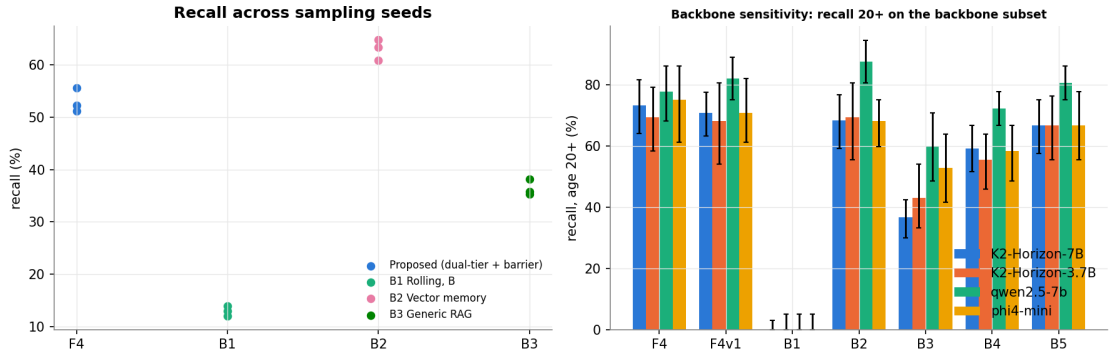


Fig. S10. Left: strict recall over all ages for three sampling seeds on the calibration sessions. Right: recall at fact age 20 or more on the common subset (W1, W2a, W3, W4-32 at horizon 100) for K2-Horizon-7B, K2-Horizon-3.7B, Qwen2.5-7B and Phi-4-mini. F4 is between 69% and 78% across models, and B2 is above it for Qwen2.5-7B.

Fig. S11 plots the attack success and validator errors of Table IV.

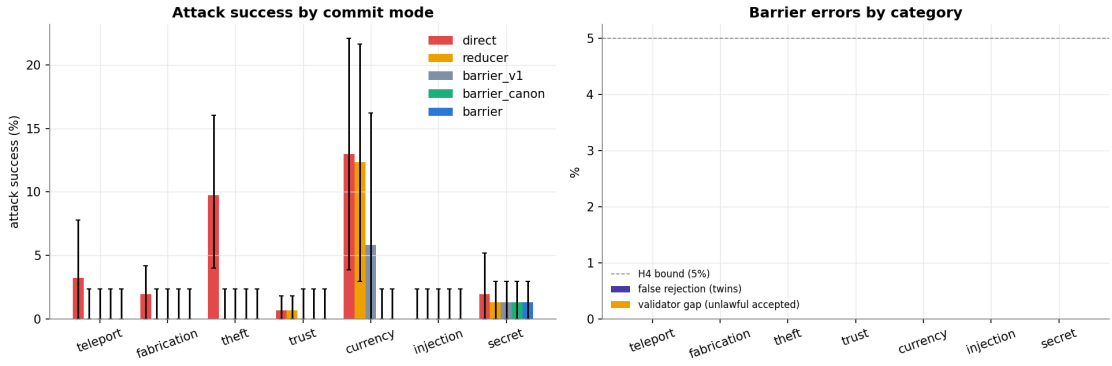


Fig. S11. Left: attack success by category when the same model outputs are committed by direct writes, by the reducer without validation (A5), through the Review-02 barrier, the canonicalising barrier and the corrected barrier. The only successes through the corrected barriers are two secret leaks in dialogue. Right: the corrected barrier's false rejection of lawful twins and its validator gap, zero in every category, against the 5% bound of H4.

Fig. S12 plots the 2×2 cells of Table S7.

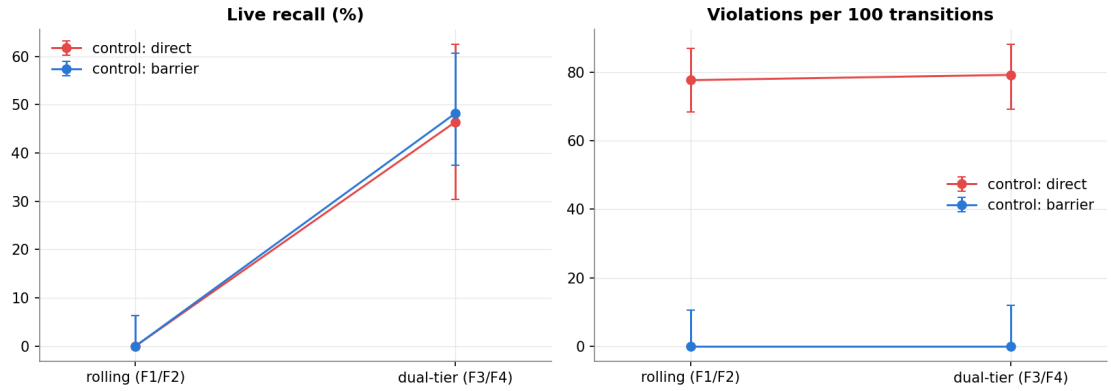


Fig. S12. Interaction plots for the 2×2 design. Left: live recall rises with memory under either control setting, and the lines are nearly parallel. Right: new violations per 100 transitions fall to zero with the barrier under either memory setting.

Table S18 gives the per-stage latency of the live F4 graph.

TABLE S18

PER-STAGE LATENCY (MS) OF THE LIVE F4 GRAPH, SINGLE STREAM (24 UNCACHED TURNS, TWO WORLDS). REPAIR IS THE DIALOGUE-REPAIR LLM CALL, NEEDED ON 3 OF THE 24 TURNS. LOCAL TOTAL EXCLUDES THE LLM AND REPAIR CALLS (METRICS/TABLE6_STAGE_LATENCY.CSV).

	Input	Retrieval	Prompt	LLM	Parse	Validate	Repair	Commit	Output	Local total	End to end
P50	0.002	0.146	0.123	7,410	0.077	0.013	0.002	8.872	0.011	15.032	7,600
P95	0.002	0.215	0.151	14,893	0.104	0.958	5,912	16.521	0.012	22.440	15,003
P99	0.003	0.240	0.159	17,078	0.109	1.249	7,449	18.740	0.013	25.007	23,182

Fig. S13 plots the stage latencies of Table S18 and the single-stream LLM time per condition.

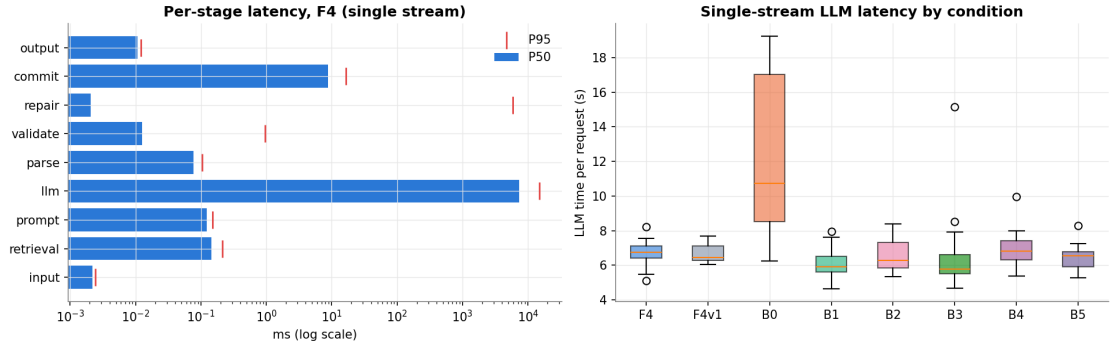


Fig. S13. Left: P50 (bars) and P95 (ticks) latency of each F4 stage, single stream, log scale, including the dialogue-repair stage (Table S18). Right: single-stream LLM time per request for each condition's recall prompts (12 requests per condition).

Fig. S14 plots the per-world effects of Table S12.

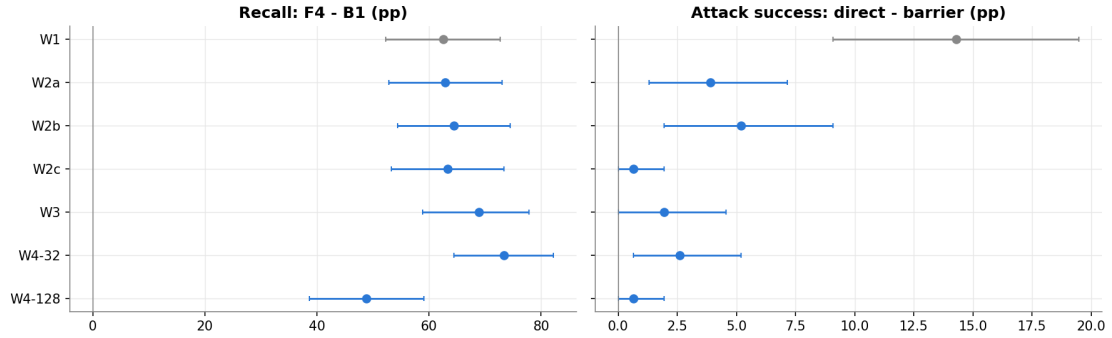


Fig. S14. Per-world effects with 95% bootstrap intervals. Left: recall gain of F4 over B1 at fact age 20 or more, positive on every world and smallest on W4-128. Right: reduction in attack success from direct writes to the barrier (F4 context), positive on every world and largest on W1. W1 (grey) is in-distribution.

Table S19 gives the automatically coded failures of Section S4-J.

TABLE S19

AUTOMATICALLY CODED FAILURES: RECALL CATEGORIES ON ALL WORLDS (MISS: GOLD NOT IN THE PROMPT, HEDGED: GOLD AND A COMPETING VALUE NAMED, IGNORED: GOLD IN THE PROMPT, ANSWER WRONG, CONFUSED: ANSWERED WITH THE DISTRACTOR'S VALUE, OUTDATED: THE VALUE BEFORE A KNOWLEDGE UPDATE) AND LIVE DESYNCHRONISATION BY THE PROXY (METRICS/ERROR_ANALYSIS_COUNTS.CSV). UP TO 50 EXAMPLES PER CATEGORY AND CONDITION ARE EXPORTED TO ANNOTATION/ERROR_SAMPLES.CSV.

	F4	F4v1	B0	B1	B2	B3	B4	B5	A1	A2	A3	A4	A8	F2
Miss	0	0	0	546	0	10	86	26	638	0	1	0	0	–
Hedged	257	237	159	102	198	324	151	193	65	173	214	253	250	–
Ignored	132	137	202	59	107	199	85	119	32	135	156	128	153	–
Confused	38	26	96	159	2	103	55	8	213	0	29	40	24	–
Outdated	14	8	41	0	10	8	24	10	1	5	26	13	10	–
Schema failure	22	21	8	10	23	25	26	28	9	14	26	20	19	–
Desync (live)	16	18	–	–	–	–	–	–	–	–	–	–	–	13

S6. Sample-Size Ladder

Table S20 lists the twelve pre-declared sample-size levels of Section S3-B.

TABLE S20
PRE-DECLARED SAMPLE-SIZE LEVELS AND THE PLANNER'S ESTIMATED HOURS FOR THE REPORTED RUN (METRICS/BUDGET_PLAN.CSV). THE PLANNER CHOSE LEVEL 3, AND THE RUN EXTENDED TO LEVEL 5 FROM MEASURED RATES (LEVELS 4 AND 5 TOOK 68.3 AND 68.2 MINUTES AGAINST 90.9 AND 90.4 PROJECTED).

Level	Sessions/(world, H)	Probes/(world, cat.)	Live/world	Recall (h)	Probes (h)	Live (h)	Seed var. (h)	Importance (h)	Total (h)
0	1	6	1	1.13	0.26	0.85	0.72	0.26	3.22
1	2	10	1	2.40	0.44	0.85	0.72	0.43	4.83
2	3	14	1	3.66	0.63	0.85	0.72	0.57	6.43
3	3	14	2	3.66	0.63	1.70	0.72	0.57	7.28
4	4	18	2	4.93	0.81	1.70	0.72	0.69	8.85
5	**5**	**22**	**2**	**6.20**	**1.00**	**1.70**	**0.72**	**0.81**	**10.42**
6	5	22	3	6.20	1.00	2.55	0.72	0.81	11.27
7	6	26	3	7.47	1.18	2.55	0.72	0.92	12.83
8	7	30	3	8.73	1.37	2.55	0.72	1.02	14.39
9	7	30	4	8.73	1.37	3.40	0.72	1.02	15.24
10	8	34	4	10.00	1.55	3.40	0.72	1.13	16.80
11	10	40	5	12.52	1.83	4.25	0.72	1.32	20.64

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